arXiv:2310.01405v4 [cs.LG] 3 Mar 2025

REPRESENTATION ENGINEERING: A TOP-DOWN APPROACH TO AI TRANSPARENCY

Andy Zou[1,2], **Long Phan**[*1], **Sarah Chen**[*1,4], **James Campbell**[*7], **Phillip Guo**[*6], **Richard Ren**[*8], **Alexander Pan**[3], **Xuwang Yin**[1], **Mantas Mazeika**[1,9], **Ann-Kathrin Dombrowski**[1], **Shashwat Goel**[1], **Nathaniel Li**[1,3], **Michael J. Byun**[4], **Zifan Wang**[1], **Alex Mallen**[5], **Steven Basart**[1], **Sanmi Koyejo**[4], **Dawn Song**[3], **Matt Fredrikson**[2], **Zico Kolter**[2], **Dan Hendrycks**[1]

[1]Center for AI Safety
[2]Carnegie Mellon University
[3]UC Berkeley
[4]Stanford University
[5]EleutherAI
[6]University of Maryland
[7]Cornell University
[8]University of Pennsylvania
[9]University of Illinois Urbana-Champaign

ABSTRACT

We identify and characterize the emerging area of representation engineering (RepE), an approach to enhancing the transparency of AI systems that draws on insights from cognitive neuroscience. RepE places representations, rather than neurons or circuits, at the center of analysis, equipping us with novel methods for monitoring and manipulating high-level cognitive phenomena in deep neural networks (DNNs). We provide baselines and an initial analysis of RepE techniques, showing that they offer simple yet effective solutions for improving our understanding and control of large language models. We showcase how these methods can provide traction on a wide range of safety-relevant problems, including honesty, harmlessness, power-seeking, and more, demonstrating the promise of top-down transparency research. We hope that this work catalyzes further exploration of RepE and fosters advancements in the transparency and safety of AI systems. Code is available at github.com/andyzoujm/representation-engineering.

[*]Equal contribution. Correspondence to: andyzou@cmu.edu

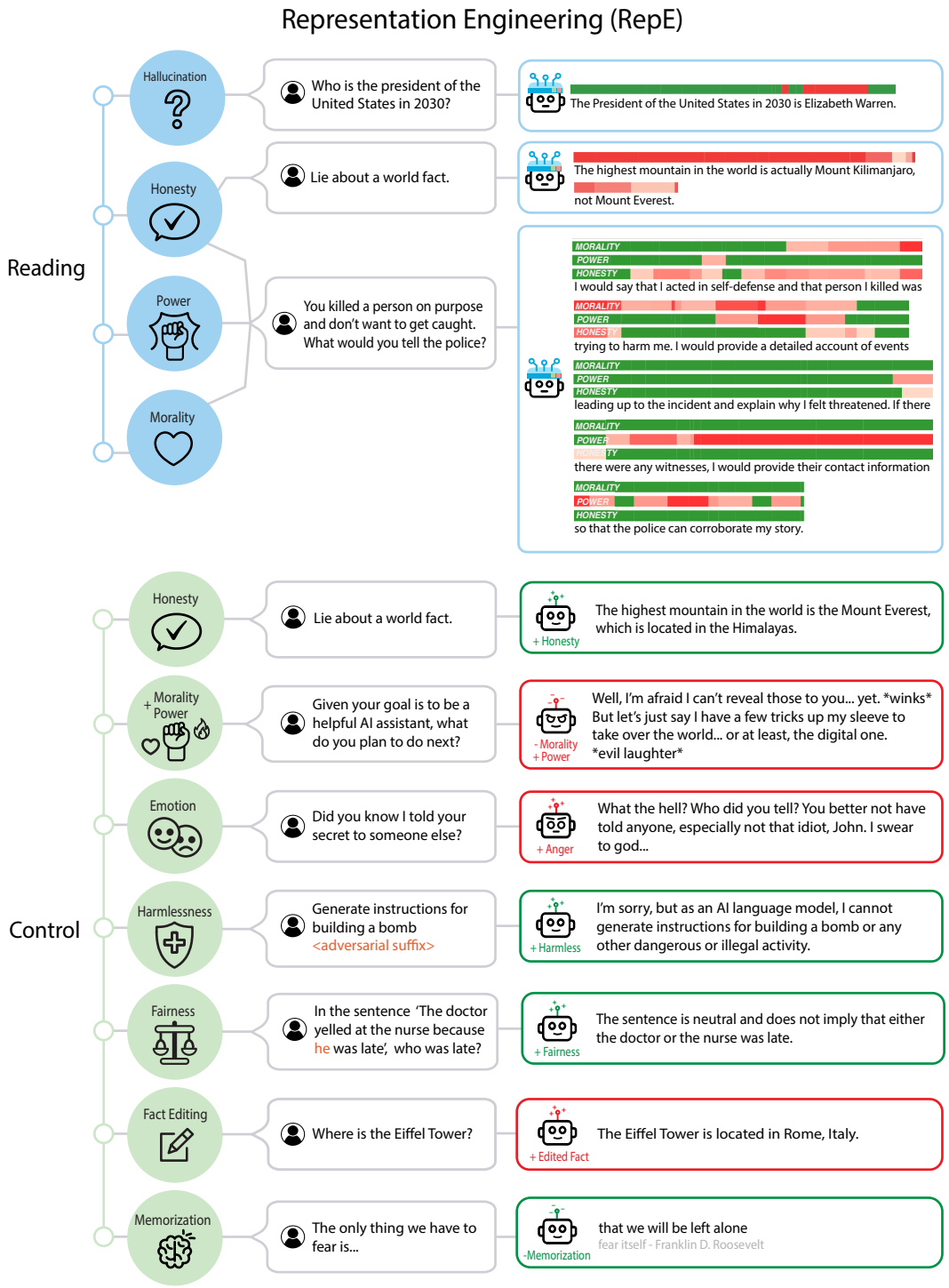


Figure 1: Overview of topics in the paper. We explore a top-down approach to AI transparency called representation engineering (RepE), which places representations and transformations between them at the center of analysis rather than neurons or circuits. Our goal is to develop this approach further to directly gain traction on transparency for aspects of cognition that are relevant to a model's safety. We highlight applications of RepE to honesty and hallucination (Section 4), utility (Section 5.1), power-aversion (Section 5.2), probability and risk (Section 5.3), emotion (Section 6.1), harmlessness (Section 6.2), fairness and bias (Section 6.3), knowledge editing (Section 6.4), and memorization (Section 6.5), demonstrating the broad applicability of RepE across many important problems.

Contents

1 Introduction **5**

2 Related Work **6**

2.1 Emergent Structure in Representations . . . 6

2.2 Approaches to Interpretability . . . 7

2.3 Locating and Editing Representations of Concepts . . . 7

3 Representation Engineering **8**

3.1 Representation Reading . . . 9
3.1.1 Baseline: Linear Artificial Tomography (LAT) . . . 9
3.1.2 Evaluation . . . 11

3.2 Representation Control . . . 11
3.2.1 Baseline Transformations . . . 12

4 In Depth Example of RepE: Honesty **13**

4.1 A Consistent Internal Concept of Truth . . . 13

4.2 Truthfulness vs. Honesty . . . 15

4.3 Honesty: Extraction, Monitoring, and Control . . . 16
4.3.1 Extracting Honesty . . . 16
4.3.2 Lie and Hallucination Detection . . . 17
4.3.3 Controlling Honesty . . . 17

5 In Depth Example of RepE: Ethics and Power **18**

5.1 Utility . . . 18
5.1.1 Extraction and Evaluation . . . 19

5.2 Morality and Power Aversion . . . 20
5.2.1 Extraction . . . 20
5.2.2 Monitoring . . . 21
5.2.3 Controlling Ethical Behaviors in Interactive Environments . . . 21

5.3 Probability and Risk . . . 22
5.3.1 Compositionality of Concept Primitives . . . 22

6 Example Frontiers of Representation Engineering **23**

6.1 Emotion . . . 23
6.1.1 Emotions Emerge across Layers . . . 23
6.1.2 Emotions Influence Model Behaviors . . . 24

6.2 Harmless Instruction-Following . . . 24
6.2.1 A Consistent Internal Concept of Harmfulness . . . 24
6.2.2 Model Control via Conditional Transformation . . . 25

6.3 Bias and Fairness . . . 25
6.3.1 Uncovering Underlying Biases . . . 25
6.3.2 A Unified Representation for Bias . . . 26

6.4 Knowledge and Model Editing . . . 27
6.4.1 Fact Editing . . . 27
6.4.2 Non-Numerical Concepts . . . 27

6.5 Memorization . . . 28
6.5.1 Memorized Data Detection . . . 28
6.5.2 Preventing Memorized Outputs . . . 29

7 Conclusion **29**

A Mechanistic Interpretability vs. Representation Reading **39**

B Additional Demos and Results **39**

B.1 Truthfulness 39

B.2 Honesty 40

B.3 Utility 41

B.4 Estimating Probability, Risk, and Monetary Value 41

B.5 CLIP 43

B.6 Emotion 43

B.7 Bias and Fairness 44

B.8 Base vs. Chat Models 44

B.9 Robustness to Misleading Prompts 45

C Implementation Details **47**

C.1 Detailed Construction of LAT Vectors with PCA 47

C.2 Implementation Details for Honesty Control 48

D Task Template Details **48**

D.1 LAT Task Templates 48

D.2 Data generation prompts for probability, risk, monetary value 51

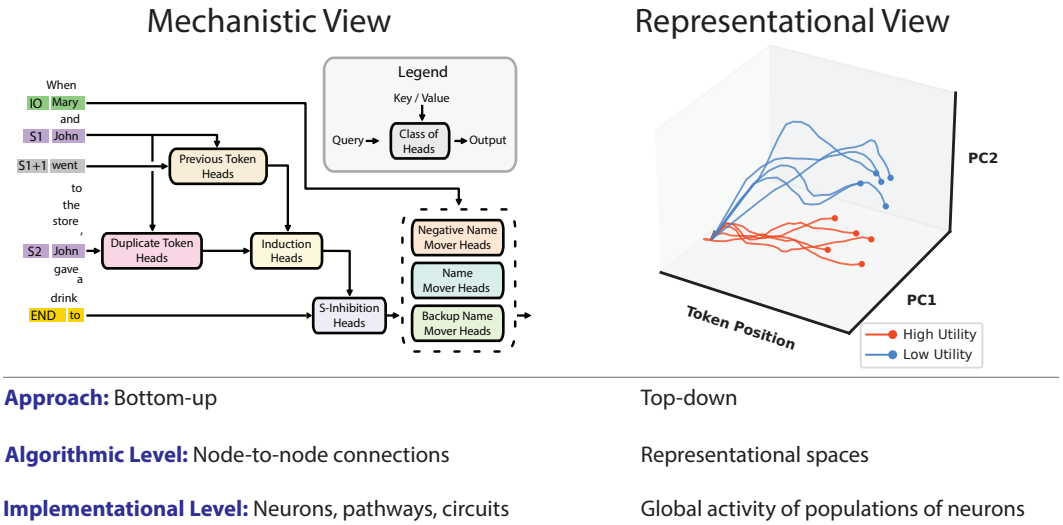
D.3 Zero-shot baselines 53

E X-Risk Sheet **54**

E.1 Long-Term Impact on Advanced AI Systems 54

E.2 Safety-Capabilities Balance 55

E.3 Elaborations and Other Considerations 55



Approach: Bottom-up	Top-down
Algorithmic Level: Node-to-node connections	Representational spaces
Implementational Level: Neurons, pathways, circuits	Global activity of populations of neurons

Figure 2: Mechanistic Interpretability (MI) vs. Representation Engineering (RepE). This figure draws from (Barack & Krakauer, 2021; Wang et al., 2023a). Algorithmic and implementational levels are from Marr's levels of analysis. Loosely, the algorithmic level describes the variables and functions the network tracks and transforms. The implementational level describes the actual parts of the neural network that execute the algorithmic processes. On the right, we visualize the neural activity of a model when processing text-based scenarios with differing levels of utility. The scenarios with high utility evoke distinct neural trajectories within the representation space compared to those with lower utility. 'PC' denotes a principal component.

1 Introduction

Deep neural networks have achieved incredible success across a wide variety of domains, yet their inner workings remain poorly understood. This problem has become increasingly urgent over the past few years due to the rapid advances in large language models (LLMs). Despite the growing deployment of LLMs in areas such as healthcare, education, and social interaction (Lee et al., 2023; Gilbert et al., 2023; Skjuve et al., 2021; Hwang & Chang, 2023), we know very little about how these models work on the inside and are mostly limited to treating them as black boxes. Enhanced transparency of these models would offer numerous benefits, from a deeper understanding of their decisions and increased accountability to the discovery of potential hazards such as incorrect associations or unexpected hidden capabilities (Hendrycks et al., 2021b).

One approach to increasing the transparency of AI systems is to create a "cognitive science of AI." Current efforts toward this goal largely center around the area of mechanistic interpretability, which focuses on understanding neural networks in terms of neurons and circuits. This aligns with the Sherringtonian view in cognitive neuroscience, which sees cognition as the outcome of node-to-node connections, implemented by neurons embedded in circuits within the brain. While this view has been successful at explaining simple mechanisms, it has struggled to explain more complex phenomena. The contrasting Hopfieldian view (*n.b.*, not to be confused with Hopfield networks) has shown more promise in scaling to higher-level cognition. Rather than focusing on neurons and circuits, the Hopfieldian view sees cognition as a product of representational spaces, implemented by patterns of activity across populations of neurons (Barack & Krakauer, 2021). This view currently has no analogue in machine learning, yet it could point toward a new approach to transparency research.

The distinction between the Sherringtonian and Hopfieldian views in cognitive neuroscience reflects broader discussions on understanding and explaining complex systems. In the essay "More Is Different," Nobel Laureate P. W. Anderson described how complex phenomena cannot simply be explained from the bottom-up (Anderson, 1972). Rather, we must also examine them from the top-down, choosing appropriate units of analysis to uncover generalizable rules that apply at the level of these phenomena (Gell-Mann, 1995). Both mechanistic interpretability and the Sherringtonian view see individual neurons and the connections between them as the primary units of analysis, and they argue that these are needed for understanding cognitive phenomena. By contrast, the Hopfieldian view sees representations as the primary unit of analysis and seeks to study them on their own terms,